Using Twos, Fives, and Tens as Anchor Facts

Skip counting by twos, fives, and tens, doubling a two-fact, building from a five-fact, taking one group away from a ten-fact, and using anchor facts to divide

The one idea behind every problem. Some facts are easy to count: twos, fives, and tens. Those are your **anchor facts**. Every harder fact sits right next door to one of them, so you never have to guess — you start at an anchor and take one small step.

One more group than the five-fact $\rightarrow$ add the group size once
Double the two-fact $\rightarrow$ the four-fact
One group less than the ten-fact $\rightarrow$ the nine-fact

Reference card. No numbers are filled in — use the numbers given in each problem.

Directions.

- Show the anchor fact you started from. That step is part of the answer here, not something to skip.
- You may draw dots, circles, or rows of counters in the work space — pictures count as work.
- Write your final answer on the answer line.

1. Trina buys 7 packs of erasers. Each pack holds 2 erasers.

Skip-count by twos to fill the blanks, then write the multiplication fact the count shows.

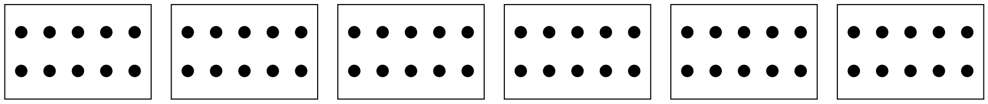
2, _____, _____, _____, _____, _____, _____

How many erasers does Trina have in all?

Answer: _________________

2. A marker rack holds 10 markers when it is full. The art table has 6 full racks.

Each rack below stands for one full rack of markers. Write the running total under each one, then write the multiplication fact.



How many markers are on the art table?

Answer: ________________

3. Nia already knows her five-facts by counting nickels, so she knows $8 \times 5 = 40$ without thinking.

Now she needs 8×6. Only one thing changed: there is *one more group of* 8.

(a) Draw 5 groups of 8 dots, then draw the one extra group in a different colour or circle it.

(b) Start at the anchor fact 40 and take that one step. What is 8×6?

Answer: ________________

4. Ravi knows the two-fact $6 \times 2 = 12$.

A four-fact is always the two-fact *doubled*, because 4 is two 2s.

(a) Draw 6 rows with 2 dots in each row. Then draw the whole picture again right beside it.

(b) Use your picture to explain in one sentence why doubling works.

(c) What is 6×4?

Answer: ___________________

5. Tens are the easiest anchor of all, and 9 is just one group short of 10.

Priya needs 7×9. She starts at the ten-fact $7 \times 10 = 70$.

(a) How many groups of 7 does she have too many? Write that number.

(b) Take that many away from 70 and write the answer to 7×9.

(c) Careful: taking away one *group* is not the same as taking away one *counter*. Say in one sentence what the difference is.

Answer: ___________________

6. A craft box holds 45 beads. Each bracelet uses 5 beads.

Anchor facts work backwards too: instead of dividing, count *up* by fives and keep track of how many jumps it takes.

5, 10, _____, _____, _____, _____, _____, _____, _____

How many bracelets can be made?

Answer: ________________

7. Sevens have no anchor of their own — so build one. A seven-fact can be split into a five-fact plus a two-fact, because $5 + 2 = 7$.

Tomas needs 7×7.

(a) Draw 7 rows of 7 dots. Draw a line that cuts every row into a part with 5 dots and a part with 2 dots.

(b) Write the five-fact for the left part and the two-fact for the right part.

(c) Add them. What is 7×7?

Answer: ________________

8. Challenge. Marco has 56 stickers. He puts 7 stickers in each row.

He does not know his seven-facts yet, but he does know the anchor fact $7 \times 5 = 35$.

(a) How many stickers are left over after those first five rows?

(b) Keep counting on by sevens from 35 and record each jump until you reach 56. How many extra rows did that take?

(c) How many rows does Marco fill altogether?

Answer: ____________________